Lezione 23 - 10/05/2024

SEGNALI		FOURIER
CONTINUI	$\longleftrightarrow$	APERIODICO
APERIODICI	$\longleftrightarrow$	CONTINUO

Facciamo un paio di esercizi prima di passare al teorema del campionamento

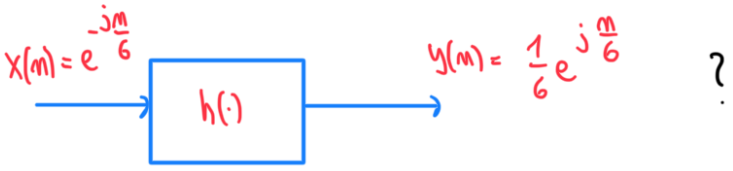
ESERCIZIO 5 - APPELLO 2 (2021)

(5.a) È VERO CHE $x * y(n) = x(n+8) * y(n-8)$

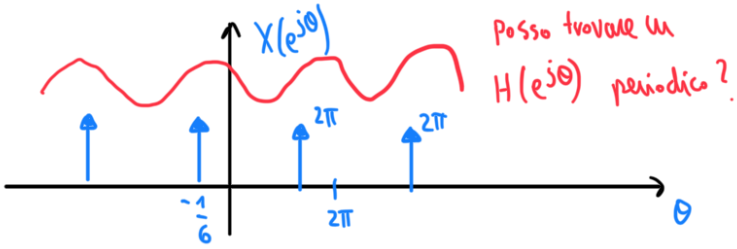
RISPOSTA: VERO, PERCHÉ LE TRASLAZIONI NELLA CONVOLUZIONE SI SOMMANO

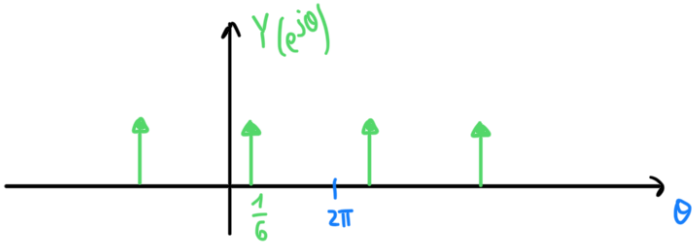
$$x(n+8) * y(n-8) = x * y(n-8+8) = x * y(n)$$

(5.b) POSSO IDENTIFICARE $h(n)$ TALE CHE

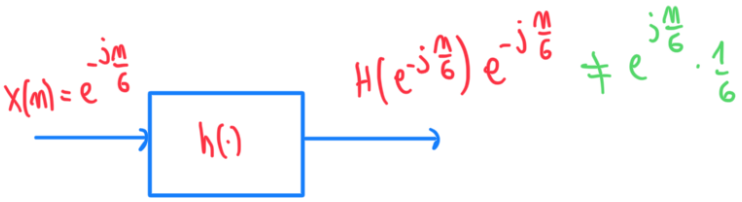


SOL. PASSIAMO NEL DOMINIO DI FOURIER





abbiamo visto che gli esponenziali complessi sono autofunzioni di filtri

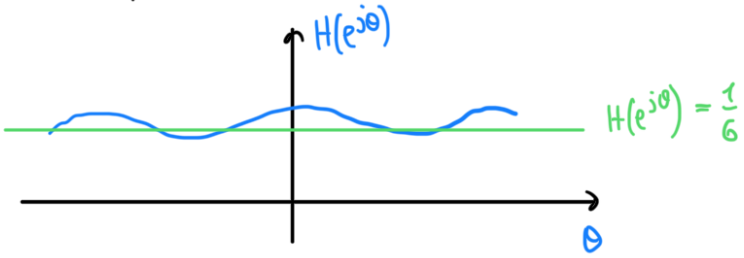


QUINDI NON PUÒ ESISTERE UN FILTRO $h(\cdot)$ COSÌ, PERCHÉ IL MASSIMO CHE PUÒ FARE È MOLTIPLICARE $e^{-j\frac{n}{6}}$ PER UNA COSTANTE $H(j\omega)$

(5.c) INVECE PUÒ ESISTERE UN FILTRO...

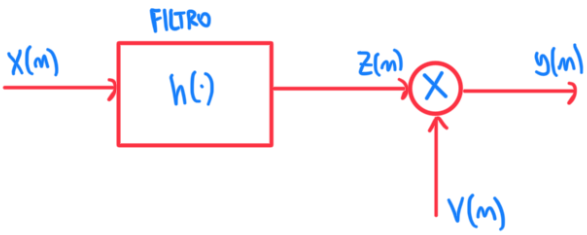
?

SÌ, BASTA CHE $h(n) = \frac{1}{6}\delta(n)$



Altro esercizio:

ESERCIZIO 3 - APPELLO 2 (2021)



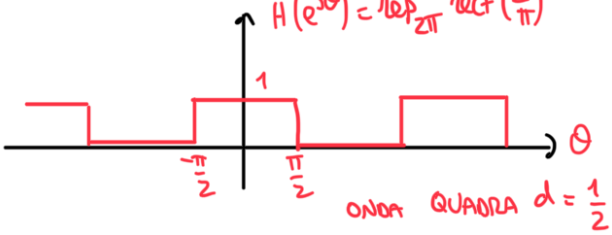
$y(n) = ?$

$X(n) = \delta(n-1) - \delta(n+1)$

FILTRO PASSA BASSO CON FASE DI TAGLIO $\frac{\pi}{2}$

$V(n) = 1 - e^{-j\pi n}$

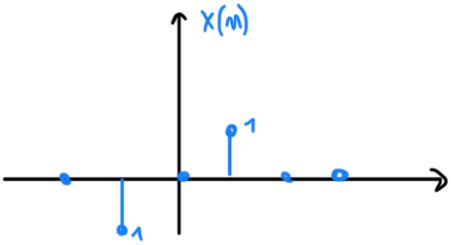
$H(e^{j\theta}) = \text{rep}_{2\pi}\, \text{rect}\left(\frac{\theta}{\pi}\right)$



Sol. NEL DOMINIO DEL TEMPO:

$$z(n) = x * h(n) = [\delta(n-1) - \delta(n+1)] * h(n)$$

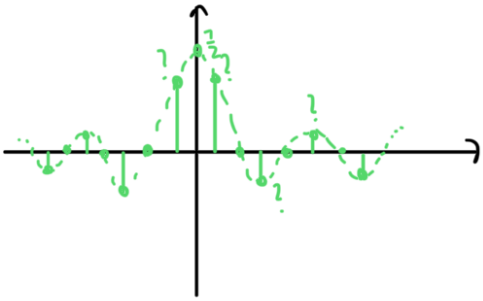
$$= h(n-1) - h(n+1)$$

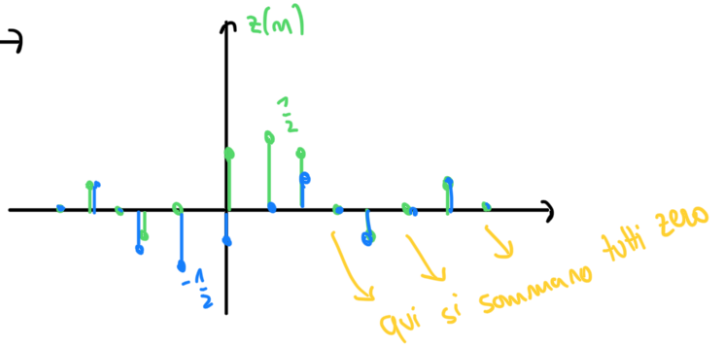


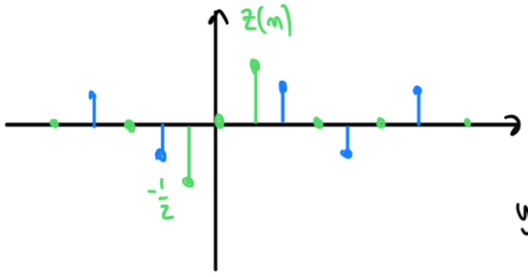
h nel tempo è un sinc campionato

$$h(n) = d\,\text{sinc}(nd) = \frac{1}{2}\text{sinc}\left(\frac{n}{2}\right)$$

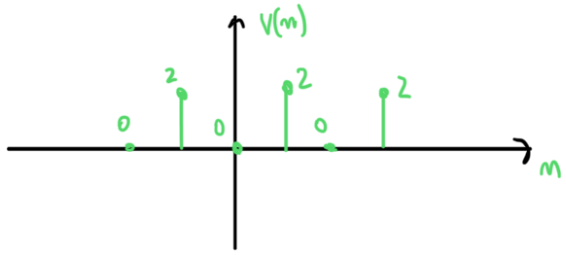
$$\frac{1}{2}\text{sinc}\left(\frac{t}{2}\right) \xrightarrow{\mathcal{F}} \frac{1}{\cancel{2}}\cdot\cancel{2}\,\text{rect}\left(\frac{\omega}{\cancel{2}\pi}\cdot\cancel{2}\right)$$



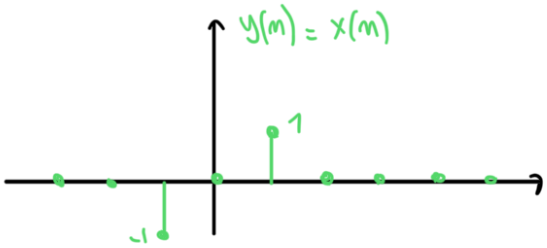




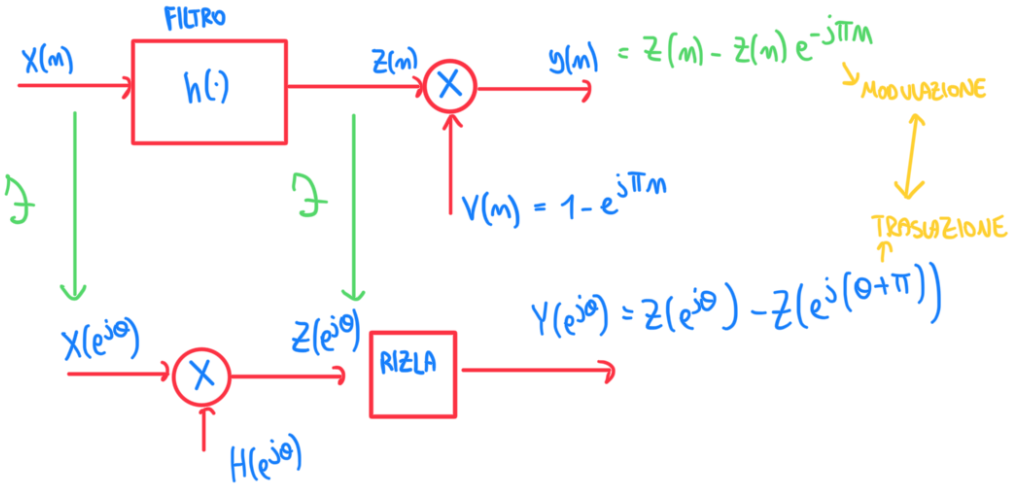
$$y(n) = V(n)z(n) = z(n)\left[1 - \overbrace{e^{-j\pi n}}^{(-1)^n}\right]$$



il prodotto sta andando a cancellare tutti i valori agli istanti pari

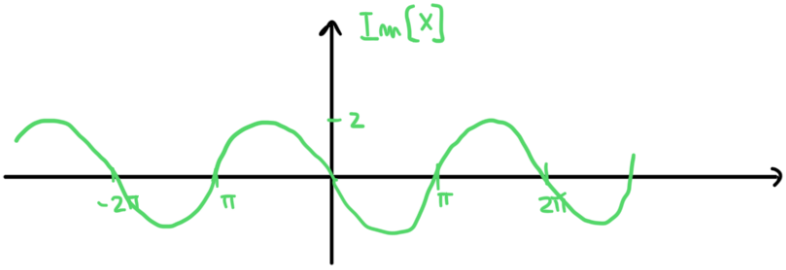


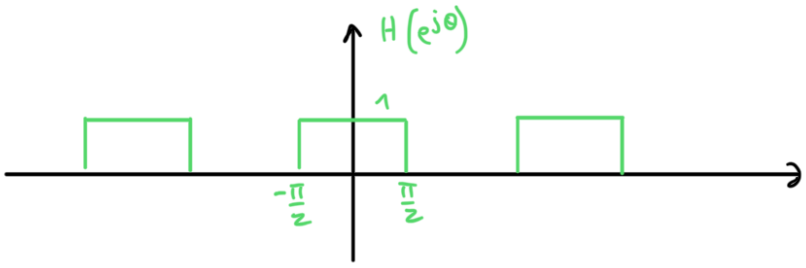
QUESTA SOLUZIONE È CORRETTA. TUTTAVIA L'ESERCIZIO SI PUÒ RISOLVERE ANCHE NEL DOMINIO DI FOURIER

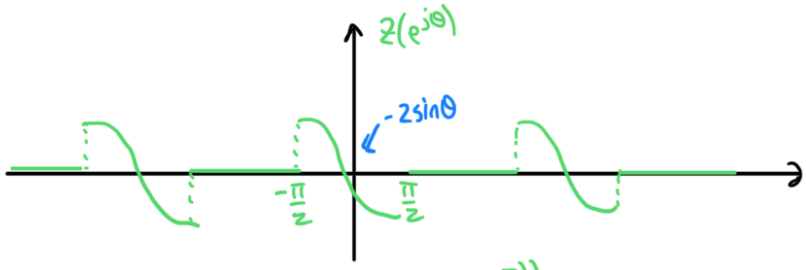


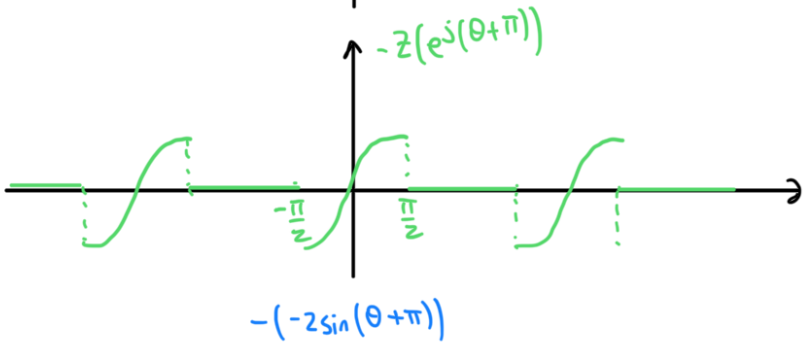
$$x(n) = \delta(n-1) - \delta(n+1)$$

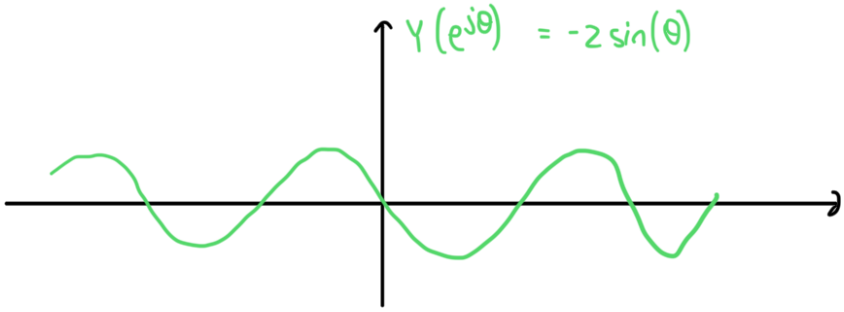
$$X(e^{j\theta}) = \sum_n x(n)\, e^{-jn\theta} = e^{-j\theta} - e^{-j\theta} = -\left(e^{j\theta} - e^{-j\theta}\right) = -2j\sin\theta$$






$Z(e^{j\theta})$
$-2\sin\theta$
$-\frac{\pi}{2}$
$\frac{\pi}{2}$


$-Z(e^{j(\theta+\pi)})$
$-\frac{\pi}{2}$
$\frac{\pi}{2}$
$-(-2\sin(\theta+\pi))$


$Y(e^{j\theta}) = -2\sin(\theta)$